

For a **homography**, the key difference from an affine transformation is that we have **8 independent parameters**, so we need **4 point matches**.

Numerical toy example

Suppose four points in image 1 correspond to four points in image 2:

Original (x, y)	Transformed (x', y')
(0, 0)	(0, 0)
(1, 0)	(2, 0)
(0, 1)	(0, 2)
(1, 1)	(2/3, 2/3)

We want to find the homography matrix

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}.$$

Because multiplying all elements of H by the same number gives the same transformation, we can set

$$h_{33} = 1.$$

So there are **8 unknowns**.

1. Start with the homography matrix

We write

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}.$$

A point in the first image is

$$p = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

The homography maps it to

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

The symbol $\sim$ means "equal up to a scale factor."

2. Multiply the matrix

Do the matrix multiplication:

$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

gives

$$\begin{bmatrix} h_{11}x + h_{12}y + h_{13} \\ h_{21}x + h_{22}y + h_{23} \\ h_{31}x + h_{32}y + h_{33} \end{bmatrix}.$$

Let's call this

$$\begin{bmatrix} X \\ Y \\ W \end{bmatrix}.$$

So:

$$X = h_{11}x + h_{12}y + h_{13}$$

$$Y = h_{21}x + h_{22}y + h_{23}$$

$$W = h_{31}x + h_{32}y + h_{33}.$$

3. Why do we divide?

This is the key idea.

In **homogeneous coordinates**, the points

$$\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

represent the **same image point**.

To convert back to ordinary (x,y) coordinates, we divide by the third coordinate W :

$$x' = \frac{X}{W}, \quad y' = \frac{Y}{W}.$$

Therefore,

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

and

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}.$$

Why is this different from an affine transformation?

For an **affine transformation**:

$$x' = ax + by + t_x$$

$$y' = cx + dy + t_y.$$

But for a homography:

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + h_{33}}$$

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + h_{33}}.$$

The denominator depends on x and y .

That dependence allows a homography to model **perspective effects**—for example, the apparent narrowing of railway tracks or a rectangular poster photographed at an angle.

Affine: linear transformation + translation

Homography: affine-like transformation **plus perspective division**

And the **third row of H** ,

$$[h_{31} \quad h_{32} \quad h_{33}],$$

is what creates that perspective effect.

If $h_{31} = h_{32} = 0$, the denominator is constant, and the homography essentially reduces to an affine transformation (after normalization).

The homography equations

For a point (x, y) ,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

This gives

$$x' = \frac{h_{11}x + h_{12}y + h_{13}}{h_{31}x + h_{32}y + 1}$$

and

$$y' = \frac{h_{21}x + h_{22}y + h_{23}}{h_{31}x + h_{32}y + 1}.$$

Use the four matches

For $(0, 0) \rightarrow (0, 0)$:

$$h_{13} = 0, \quad h_{23} = 0.$$

For $(1, 0) \rightarrow (2, 0)$:

$$h_{11} = 2, \quad h_{21} = 0.$$

For $(0, 1) \rightarrow (0, 2)$:

$$h_{12} = 0, \quad h_{22} = 2.$$

Now we only need h_{31} and h_{32} .

For the final correspondence

$$(1,1) \rightarrow (2/3, 2/3),$$

the x -equation is

$$\frac{2}{h_{31} + h_{32} + 1} = \frac{2}{3}.$$

Therefore

$$h_{31} + h_{32} + 1 = 3$$

and hence

$$h_{31} + h_{32} = 2.$$

For simplicity, choose

$$h_{31} = 1, \quad h_{32} = 1.$$

So the homography is

$$H = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Check the last point

Take (1,1):

$$H \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Because homogeneous coordinates are normalized by the last coordinate,

$$\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2/3 \\ 2/3 \\ 1 \end{bmatrix}.$$

So indeed

$$(1, 1) \rightarrow (2/3, 2/3).$$

Homography has 8 independent parameters

Each point correspondence provides **2 equations**:

$$(x, y) \rightarrow (x', y') \Rightarrow 2 \text{ equations.}$$

Therefore,

$$\frac{8}{2} = 4 \text{ point matches}$$

are needed to solve a homography.

Important: the 4 points should be in a non-degenerate configuration (e.g., the corners of a quadrilateral), not all on one line.